# BOQ mapping and quantity validation report

**DEMONSTRATION EXPORT - NOT SUBMISSION READY - NOT APPROVED FOR ANY REAL TENDER | approvals by: test principals in environment 'test'**

**SYNTHETIC TEST DATA - FICTIONAL CUSTOMER - NOT A REAL TENDER - NOT A MARKET QUOTE - NOT FOR SUBMISSION**

| | |
|---|---|
| Project ID | PRJ-9101 |
| Project name | Dammam Industrial Utilities - Synthetic Tender Closure |
| Customer | Gulf Process Utilities Co. (FICTIONAL SYNTHETIC CUSTOMER) |
| Document ID | PRJ-9101-D34 |
| Title | BOQ mapping and quantity validation report |
| Revision | 2 |
| Issue date | 2027-02-15 |
| Status | Preliminary (DEMONSTRATION) |
| Author | test-estimator |
| Reviewer | not yet reviewed |
| Estimate version | v2 |
| Estimate fingerprint | 48d0f9242dc910b7 |
| Template | T-BOQ-VALIDATION@1.0.0 |
| Confidentiality | CONFIDENTIAL (synthetic test data) - no real customer or supplier data |

## Summary

| Measure | Count |
|---|---|
| BOQ items | 44 |
| Unmapped | 1 |
| Unit not ok | 0 |
| Needs review | 0 |
| Possible duplicate descriptions | 1 |

## Items

| Item | Description | Unit (original) | Unit | Unit status | Qty | Mapping | Pricing | Validation |
|---|---|---|---|---|---|---|---|---|
| C-001 | Mobilisation and demobilisation | ls | ls | ok | 1 | Approved | assembly | OK |
| C-002 | Site management and supervision (monthly) | month | month | ok | 5 | Approved | assembly | OK |

| Item | Description | Unit (original) | Unit | Unit status | Qty | Mapping | Pricing | Validation |
|---|---|---|---|---|---|---|---|---|
| C-003 | Temporary facilities and site utilities | ls | ls | ok | 1 | Approved | assembly | OK |
| C-004 | HSE provisions and PPE | ls | ls | ok | 1 | Approved | assembly | OK |
| C-010 | Excavation for pipe trench | m3 | m3 | ok | 850 | Approved | assembly | OK |
| C-011 | Backfill with approved material | m3 | m3 | ok | 640 | Approved | assembly | OK |
| C-012 | Pipe trench shoring | m | m | ok | 400 | Approved | assembly | OK |
| C-013 | Excavation in hard material (quantity approximate, as required) | m3 | m3 | ok | 120 | Approved | assembly | OK |
| C-014 | Road reinstatement | m2 | m2 | ok | 180 | Approved | assembly | OK |
| C-020 | Pipe-rack foundations RC | m3 | m3 | ok | 96 | Approved | assembly | OK |
| C-021 | Formwork to pipe-rack foundations | m2 | m2 | ok | 410 | Approved | assembly | OK |
| C-022 | Reinforcement steel supply and fix | kg | kg | ok | 11500 | Approved | assembly | OK |
| C-023 | Pipe-rack structural steel supply and erect | kg | kg | ok | 28000 | Approved | assembly | OK |
| C-024 | Anchor bolts and base plates | nr | nr | ok | 120 | Approved | assembly | OK |
| C-025 | Concrete equipment plinths | m3 | m3 | ok | 14 | Approved | assembly | OK |
| C-030 | Carbon steel pipe DN100 supply | m | m | ok | 520 | Approved | assembly | OK |
| C-031 | Carbon steel pipe DN50 supply | m | m | ok | 310 | Approved | assembly | OK |
| C-032 | Pipe fabrication and installation DN100 | inch_dia | inch_dia | ok | 5200 | Approved | assembly | OK |
| C-033 | Pipe fabrication and installation DN50 | inch_dia | inch_dia | ok | 1550 | Approved | assembly | OK |
| C-034 | Gate valve DN100 supply | nr | nr | ok | 14 | Approved | assembly | OK |
| C-035 | Check valve DN100 supply | nr | nr | ok | 8 | Approved | assembly | OK |

| Item | Description | Unit (original) | Unit | Unit status | Qty | Mapping | Pricing | Validation |
|---|---|---|---|---|---|---|---|---|
| C-036 | Flange, gasket and bolt sets | set | set | ok | 96 | Approved | assembly | OK |
| C-037 | Butt welding of pipe joints | joint | joint | ok | 640 | Approved | assembly | OK |
| C-038 | Weld radiography (sample) | joint | joint | ok | 130 | Approved | assembly | OK |
| C-039 | Hydrostatic test and flushing of pipework | nr | nr | ok | 6 | Approved | assembly | OK |
| C-040 | Pipe supports, fabricated steel | kg | kg | ok | 6400 | Approved | assembly | OK |
| C-041 | Corrosion protection coating system | m2 | m2 | ok | 1900 | Approved | assembly | OK |
| C-045 | Transfer pump skid supply (long lead) | nr | nr | ok | 2 | Approved | assembly | OK |
| C-046 | Stainless steel pipe DN100 supply | m | m | ok | 80 | Approved | assembly | OK |
| C-047 | Pneumatic control valve DN100 supply | nr | nr | ok | 4 | Approved | assembly | OK |
| C-050 | Mobile crane 50t hire (BOQ days) | day | day | ok | 10 | Approved | assembly | OK |
| C-051 | Telehandler hire (BOQ days) | day | day | ok | 54 | Approved | assembly | OK |
| C-052 | Scaffolding access hire | week | week | ok | 6 | Approved | assembly | OK |
| C-055 | Fire water tie-in to existing ring main by others (interface to be confirmed) | nr | nr | ok | 3 | Approved | assembly | OK |
| C-056 | Fire hydrant assembly | nr | nr | ok | 4 | Approved | assembly | OK |
| C-060 | LV distribution board | nr | nr | ok | 3 | Approved | assembly | OK |
| C-061 | LV armoured power cable supply and install | m | m | ok | 2400 | Approved | assembly | OK |
| C-062 | Cable tray and supports | m | m | ok | 900 | Approved | assembly | OK |
| C-063 | Earthing and lightning protection | ls | ls | ok | 1 | Approved | assembly | OK |
| C-064 | Lighting poles and luminaires | nr | nr | ok | 24 | Approved | assembly | OK |
| C-070 | Fibre optic backbone | m | m | ok | 1200 | Approved | assembly | OK |

| Item | Description | Unit (original) | Unit | Unit status | Qty | Mapping | Pricing | Validation |
|---|---|---|---|---|---|---|---|---|
| C-071 | CCTV cameras | nr | nr | ok | 14 | Approved | assembly | OK |
| C-071#2 | CCTV cameras | nr | nr | ok | 14 | Unmapped | excluded | OK |
| C-072 | Access control readers | nr | nr | ok | 8 | Approved | assembly | OK |

## Assumptions, exclusions and limitations

• Quantities are human-measured BOQ quantities; the application does not measure drawings.

• Figures are taken from frozen estimate version v2 (fingerprint 48d0f9242dc910b7).

• SYNTHETIC / DEMONSTRATION data: rates, quotations, productivity norms, customer and responses are fictional test inputs, not market evidence.

## Source references

• Frozen estimate version v2 (fingerprint 48d0f9242dc910b7), totals and pricing schedule

• Registers in the application database (see document D33 for received documents, D08 for assumptions, D25 for queries)